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Development of a Tsai-Wu based Failure Model for Composite Structures for OpenAeroStruct

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Overview

With the onset of composite materials in the aerospace industry, it has become a necessity for trust worthy design and analysis models to account for the accurate modeling of the behaviour of composite materials under various operational conditions. Not only is it important to model their an-isotropic behaviour for correctly modeling the elasticity of the structures, but also their failure for accurate design engineering. TACS (Toolkit for the Analysis of Composite Structures) is a tool used for high fidelity Finite Element Analysis of composite structures, but the same is still a dire necessity for low-fidelity tools like OpenAeroStruct. OpenAeroStruct is a low-fidelity and open-source tool for low-fidelity aerostructural optimization that couples a VLM code to an FEM code that uses spatial beam elements. It incorporates the use of vonmises stress in its elements to predict the failure of the material at its critical points to analyze the failure criteria for structures during design optimization. Although it is fairly accurate as a low-fidelity tool for structures made out of isotropic materials, it is handicapped when it comes to the failure prediction of fiber-reinforced composites, which are anisotropic in nature. This paper marks the development of a failure prediction model for fiber-reinforced composite materials, which is coupled with the existing structural model of OpenAeroStruct to accurately predict the failure in FRP composites along with the consideration for a Factor of Safety, which is a necessity for design engineering. This model is then utilized to replicate a high-fidelity aerostructural optimization benchmark study by Christison Gray and Martins (2024). to gauge the capabilities of low-fidelity models equipped with the right fundamental tools.

Tsai-Wu Failure Criterion

(a) Theory

The Tsai-Wu failure criterion is a widely used theoretical approach used to predict the failure of composite materials under complex loading conditions and is particularly useful for modeling fiber reinforced composites. The Tsai-Wu criterion most accurately accounts for the anisotropic behaviour of fiber reinforced composites, even when loaded under multiple stress components. The Tsai-Wu failure envelope is defined as:

$$F_{11}\sigma_1^2 + F_{22}\sigma_2^2 + F_{66}\sigma_6^2 + F_1\sigma_1 + F_2\sigma_2^2 + 2F_{12}\sigma_1\sigma_2 \leq 1 \tag{1}$$

Where $F_{11}, F_{22}, F_{66}, F_1, F_2, F_{12}$, are determined from uniaxial and biaxial testing of unidirectional plies of the composite material, and the σ values are the axial and shear stresses in different axes.

(b) Safety Factor for Tsai-Wu Criterion

While the Tsai-Wu criterion is an accurate model for failure of FRP composites, it is necessary to implement a factor of safety while working on its applications in the real world. Unlike the division of the FOS in the yield strength of the material that is done when using vonmises stresses, the Tsai-Wu failure envelope does not scale linearly in the same fashion and is hence unable to incorporate the FOS in such a direct manner. Due to this reason, a new quantity called **Strength Ratio** was developed after the onset of composite materials in the design engineering world. Strength ratios are defined mathematically by using the constituent elements of the Tsai-Wu criterion and arranging them in such a way that they are scalable for the application of a Factor of Safety for design of composite structures. The strength ratio (SR) is defined as:

$$SR = \frac{1}{2}(a + \sqrt{a^2 + 4b}) \tag{2}$$

where

$$a = F_1\sigma_1 + F_2\sigma_2^2 \tag{3}$$
$$b = F_{11}\sigma_1^2 + F_{22}\sigma_2^2 + F_{66}\sigma_6^2 \tag{4}$$

and the SR being scalable for Factor of Safety, the Failure criteria hence becomes:

$$\frac{1}{2}(F_1\sigma_1 + F_2\sigma_2^2 + \sqrt{F_1\sigma_1 + F_2\sigma_2^{2^2} + 4F_{11}\sigma_1^2 + F_{22}\sigma_2^2 + F_{66}\sigma_6^2}) \leq \frac{1}{FOS} \tag{5}$$

OpenAeroStruct Wingbox Model

(a) Overview of the Model

The wingbox model in OpenAeroStruct primarily resembles a beam of varied sizes that are designed according to the chord length of each section of the wing. The Figure 1 shows the layout of the wingbox acorss the span of the wing.

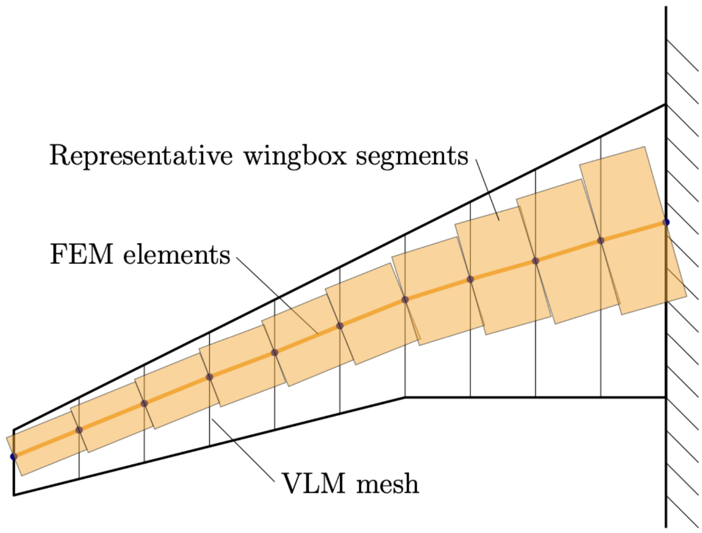


Figure 1: Wingbox layout in OpenAeroStruct, by Chauhan and Martins (2018)

(b) Computation of the failure limit

The wingbox is divided into different elements according to the meshing of the wing planform. The stresses in the wingbox elements are brought down to 4 critical points (which undergo the maximum stress due to their position and the nature of the forces that the wing undergoes during flight) and the stresses at these critical points are analyzed for determining the failure criteria of the wing structure. Figure 2 shows these critical points:

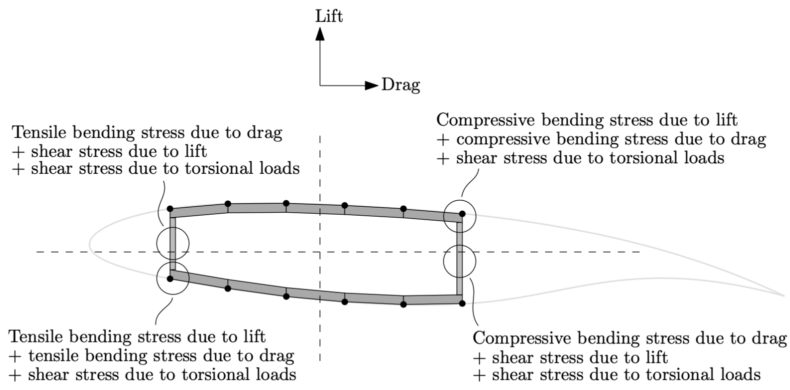


Figure 2: Critical points under consideration for maximum stress in the OpenAeroStruct wingbox model, by Chauhan and Martins (2018)

The various stresses acting at each of these points are used to find the equivalent **vonmises stresses** which are given by:

$$\sigma_{\mathrm{vm}} = \sqrt{\frac{(\sigma_x-\sigma_y)^2+(\sigma_y-\sigma_z)^2+(\sigma_z-\sigma_x)^2+6(\tau_{xy}^2+\tau_{yz}^2+\tau_{zx}^2)}{2}} \tag{6}$$

acting at each point. The vonmises stresses across each critical point, for each beam element are combined using the KS (Kreisselmeier–Steinhauser) Aggregate function in order to get a smooth gradient for gradient-based optimization and are compared with the yield strength to determine the failure criteria of the structure. Since the model uses vonmises stresses, its application is limited to the prediciton of failure in isotropic materials and paves way for the development of a failure model that can predict the same for anisotropic materials like the Carbon Fiber Reinforced Polymer composites.

Implementation of the Wingbox Model for Composite Materials

(a) Material Constitution

The constituent material of the wingbox panels comprising of the skin and spar panels are assumed to be CRFP composites. The constituent plies are assumed to be in the orientations [0°,90°,45°,−45°] with the ply fractions being [44.41%, 22.2%, 22.2%, 11.19%] and [10%, 35%, 35%, 20%] for the skin and spar elements respectively. The proprieties of the assumed CFRP composites are given in table 1:

Table 1: Composite ply properties

Property	Description	Value
E_{11}	Fiber direction modulus	117.7 GPa
E_{22}	Transverse modulus	9.7 GPa
G_{12}	In-plane shear modulus	4.8 GPa
G_{13}	Transverse shear modulus	4.8 GPa
G_{23}	Transverse shear modulus	4.8 GPa
T_1	Fiber direction tensile strength	1648 MPa
C_1	Fiber direction compressive strength	1034 MPa
T_2	Transverse tensile strength	64 MPa
C_2	Transverse compressive strength	228 MPa
S_{12}	Shear strength	71 MPa
ν_{12}	Major Poisson's ratio	0.35
ρ	Density	1550 kg/m^3

(b) Approximation of the Moduli of Elasticity

Currently, the moduli of elasticity of the entire FEM spatial beam model are assumed to be isotropic in 2D plane so as to not change the entire model and is left for the future works. The values of the moduli of elasticity are found using:

The unidirectional ply properties given in table 1 are used to find the stiffness matrix of the plies:

$$\mathbf{Q} = \begin{pmatrix} \frac{E_1}{1-\nu_{12}\nu_{21}} & \frac{\nu_{12}E_2}{1-\nu_{12}\nu_{21}} & 0 \\ \frac{\nu_{21}E_1}{1-\nu_{12}\nu_{21}} & \frac{E_2}{1-\nu_{12}\nu_{21}} & 0 \\ 0 & 0 & G_{12} \end{pmatrix}$$

The transformed reduced stiffness matrix $\bar{Q}$ for each ply is defined using:

$$[Q] = [T]^{-1}[Q][T]^{-T}$$

where T is the transformation matrix for ply orientation θ defined as:

$$[T] = \begin{pmatrix} \cos^2\theta & \sin^2\theta & 2\sin\theta\cos\theta \\ \sin^2\theta & \cos^2\theta & -2\sin\theta\cos\theta \\ -\sin\theta\cos\theta & \sin\theta\cos\theta & \cos^2\theta - \sin^2\theta \end{pmatrix}$$

The effective $\bar{Q}$ matrix for the skin and spar elements are found using a weighted mean of their respective ply fractions constitution:

$$\bar{Q_{eff\,skin}} = 44.41\%\bar{Q}_1 + 22.2\%\bar{Q}_2 + 22.2\%\bar{Q}_3 + 11.19\%\bar{Q}_4 \tag{7}$$

$$\bar{Q_{eff\,spar}} = 10\%\bar{Q}_1 + 35\%\bar{Q}_2 + 35\%\bar{Q}_3 + 20\%\bar{Q}_4 \tag{8}$$

The respective transformed reduced stiffness compliance matrices $\bar{S}_{eff}$ are found by inverting the stiffness matrices:

$$\bar{S_{eff\,skin}} = [\bar{Q_{eff\,skin}}]^{-1} \tag{9}$$

$$\bar{S_{eff\,spar}} = [\bar{Q_{eff\,spar}}]^{-1} \tag{10}$$

These compliance matrices are then used to find the approximate moduli of elasticity of the skin and spar elements using:

$$E_{11} = \frac{1}{S_{1,1}} \tag{11}$$

$$G = \frac{1}{S_{3,3}} \tag{12}$$

The moduli of skin panels are then used to model the overall values of **E** and **G** of the FEM structural model where:

$$E = 62.537GPa \tag{13}$$
$$G = 29.717GPa \tag{14}$$

(c) Computation of the Strength Ratios

Similar to the vonmises model, the 4 critical points aforementioned (fig. 2 are analyzed to find the different strains (since the stresses for each constituent ply will be depending on its orientation) acting on these points. The various strain components are namely the axial strain, torional shear strain, top bending strain, bottom bending strain, front bending strain, rear bending strain and the vertical shear strain which are found using the nodal displacements and the panel thicknesses of the wingbox elements. These strain values are categorized in the vertical and horizontal axial strains and the shear strains acting on each of the 4 critical points, as described in figure 3:

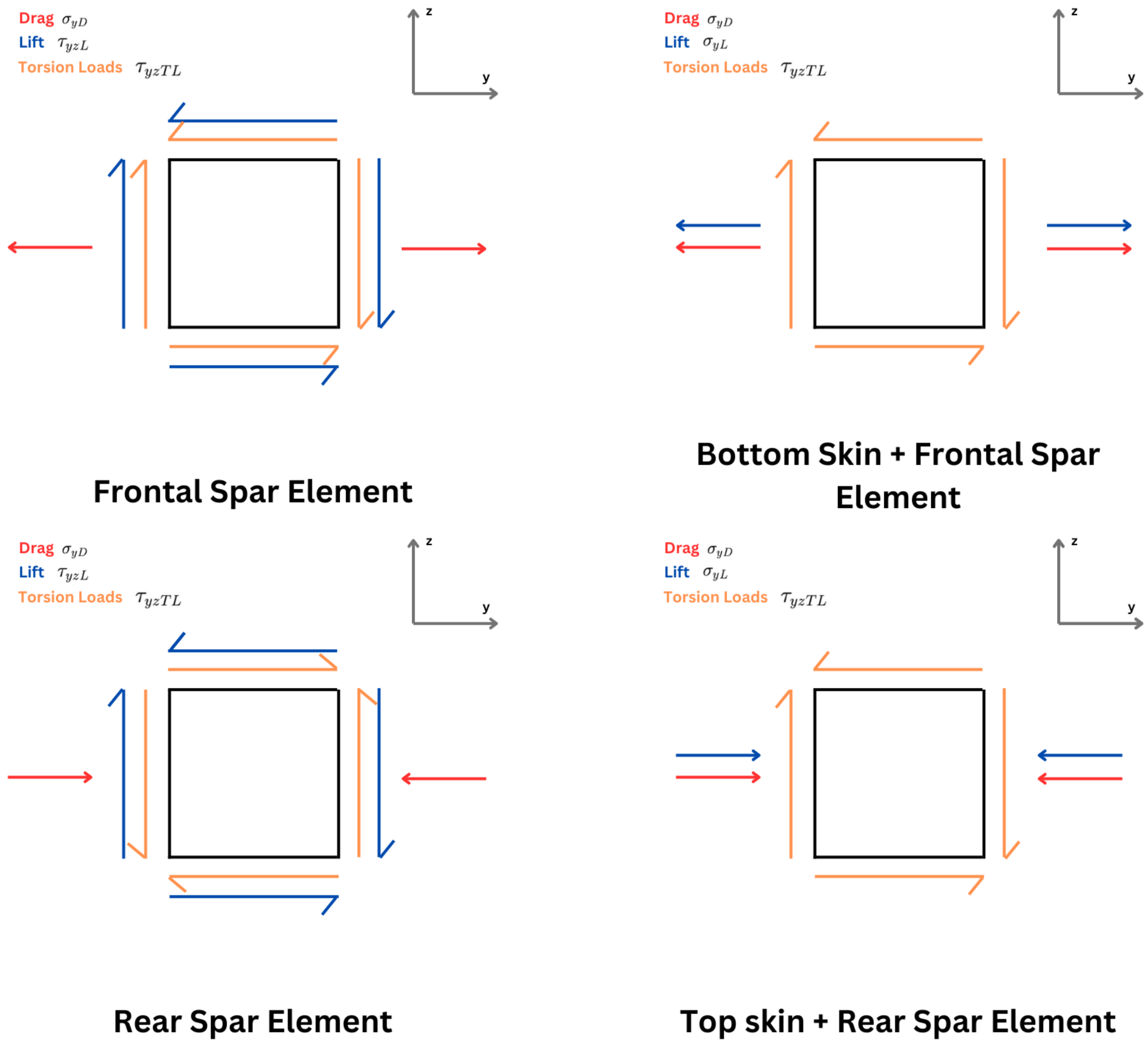


Figure 3: 2D Elements at each critical point along with the acting stresses

Then, for each critical point, the resultant strains in the ply orientations are found for each ply using the transformation:

$$\begin{pmatrix} \epsilon_1 \\ \epsilon_2 \\ \gamma_{12} \end{pmatrix} = [T] \begin{pmatrix} \epsilon_x \\ \epsilon_y \\ \gamma_{xy} \end{pmatrix}$$

The corresponding local stresses are found for each ply using:

$$\begin{pmatrix} \sigma_1 \\ \sigma_2 \\ \tau_{12} \end{pmatrix} = [Q] \begin{pmatrix} \epsilon_1 \\ \epsilon_2 \\ \gamma_{12} \end{pmatrix}$$

These local axial and shear stresses are then utilized to calculate the value of the **Strength Ratios** for each ply using Equation 5, where the coefficients are defined by:

$$F_{11} = \frac{1}{S_L^{(+)} S_L^{(-)}} \quad \text{and} \quad F_1 = \frac{1}{S_L^{(+)}} - \frac{1}{S_L^{(-)}}$$

$$F_{22} = \frac{1}{S_T^{(+)} S_T^{(-)}} \quad \text{and} \quad F_2 = \frac{1}{S_T^{(+)}} - \frac{1}{S_T^{(-)}}$$

$$F_{66} = \frac{1}{2S_{LT}^2}$$

where $S_L^{(+)}$ and $S_L^{(-)}$ are the longitudinal strengths in tension and compression respectively, $S_T^{(+)}$ and $S_T^{(-)}$ are the transverse strengths in tension and compression respectively and $S_{LT}^{(+)}$ is the shear strength of a ply.

The strength ratio values hence calculated for each ply (4 total) at each critical point (4 total), (hence 16 strength ratio values for each beam element) for all beam elements are aggregated using a **KS Aggregate** function:

$$\hat{g}_{KS}(\rho, g) = \max_j g_j + \frac{1}{\rho} \ln \left(\sum_{j=1}^{n_g} \exp \left(\rho (g_j - \max_j g_j) \right) \right). \tag{15}$$

where $\mathbf{g}$ is $(\frac{SR}{SR_{lim}} - 1)$ value for each ply and SR_{lim} is defined as:

$$SR_{lim} = \frac{1}{FOS} \tag{16}$$

The failure is determined by the value of $\max_j g_j + \hat{g}_{KS}(\rho, g)$ exceeding 0.

Aerostructural Benchmark Problem

(a) Overview

Recently, Christison Gray and Martins (2024) proposed a benchmark study for Practical Aeroelastic Optimization of Aircraft wings as a means to develope a benchmark model that is more accessible to different groups while remaining relevant to industrial use cases for modern transport aircraft design. Although this model is intended as a more accessible model, it is still intended to require the use of high-fidelity coupled aeroelastic analysis. Specifically, the use of RANS CFD for cruise drag prediction, and a built-up FE wingbox model as opposed to a condensed beam representation.
This model is implemented in this study not only as a use case for the composite wing box model, but also as a means to analyze the behaviour of low fidelity aerostructural models (like OpenAeroStruct

that couples a VLM code to an FEM code that uses spatial beam element) when served with a complicated multi-point optimization problem and to decipher the discrepencies between the models of different fidelity to mark the importance of certain aspects of the higher fidelity models to guide us in the development of better and more accurate low-fidelity tools for the wider community.
The benchmark model is designed to have 3 different cases, where the design freedom is increased from case 1 to 3. Here we only undertake the cases 1 and 2.

(b) Optimization Problem

The objective function to be minimized in Case 1 is the wingbox mass, computed from the FE model. The objective function for Case 2 is the fuel burn over a given mission. The fuel burn is computed using a two-stage process that accounts for the fuel burn in both cruise and climb. This process starts by computing the landing gross mass (LGM):

$$LGM = M_{payload} + M_{frame} + M_{fuel,res} + 2M_{wing} \tag{17}$$

The total mass of a single wing is computed using the regression model:

$$M_{wing} = 10.147 W_{wingbox}^{0.8162} \tag{18}$$

where $M_{wingbox}$ is the wingbox mass. Assuming the fuel burn during descent and landing is negligible, the mass at the start of the cruise phase, and then the takeoff gross mass (TOGM) are computed by rearranging the Breguet range equation:

$$M_{\text{cruise, start}} = LGM \exp\left(\frac{R \times TSFC}{V_{\text{cruise}}}\left(\frac{D_{\text{cruise}}}{L_{\text{cruise}}}\right)\right) \tag{19}$$

$$TOGM = M_{\text{cruise, start}} \exp\left(\frac{R_{\text{climb}} \times TSFC}{V_{\text{climb}}}\left(\frac{\cos(\gamma)}{L_{\text{cruise}}/D_{\text{cruise}}} + \sin(\gamma)\right)\right) \tag{20}$$

$$FB = TOGM - LGM \tag{21}$$

The lift and drag in the cruise condition are computed using an aeroelastic analysis, the values are doubled to get the full aircraft values, and the drag of un-modeled components (fuselage, tail, and nacelles) is added:

$$L_{cruise} = 2L_{wing} \tag{22}$$

$$D_{cruise} = 2(D_{wing} + \frac{1}{2} q_{cruise} S C_{D,frame} \tag{23}$$

where $C_{D,frame}$ is estimated using a conceptual drag build up Christison Gray and Martins (2024). The flight conditions are defined as:

Table 2: Flight conditions

Flight point	**Altitude**	**Mach Number**	**Load factor**	**Aircraft mass**
Cruise	10,400 m	0.77	1	$\sqrt{M_{\text{cruise, start}} \times LGM}$
Pull-up Maneuver	0 m	0.458	2.5	LGM
Push-down Maneuver	0 m	0.458	-1	LGM

The Boeing 717 specifications used for the aircraft are as follows:

Table 3: Boeing 717 specifications assumed in this work

Quantity	**Description**	**Value**
	Baseline wing geometry	
b	Semispan	14 m
C_{root}	Root chord	5 m
C_{tip}	Tip chord	1.5 m
S	Planform area (single wing)	45.5 m^2
MAC	Mean aerodynamic chord	3.25 m
	Masses	
M_{payload}	Payload mass	14.5×10^3 kg
M_{frame}	Operating empty mass (minus wing)	25×10^3 kg
$M_{\text{fuel, res}}$	Reserve fuel mass	2×10^3 kg
	Fuelburn calculation parameters	
R	Nominal range	3815 km
R_{climb}	Climb segment range	290 km
V_{climb}	Average climb speed	350 mph
$C_{D,\text{frame}}$	Airframe drag coefficient (fuselage + tail + nacelle)	0.0152
k_{tank}	Assumed fraction of wingbox that can store fuel	0.85
V_{aux}	Auxiliary fuel tank volume	2.763 m^3
$TSFC$	Thrust specific fuel consumption	18×10^{-6} kg N^{-1} s^{-1}
ρ_{fuel}	Fuel density	804 kg m^{-3}

The design variables for both the cases are given as:

Table 4: Design variables used in the three cases

Variable	**Description**	**Bounds**	**Case 1**	**Case 2**
x_{struct}	Structural sizing[a]	–	✓	✓
$\alpha_{2.5g}$	Pull-up maneuver angle of attack	$[-10, 20]^\circ$	✓	✓
α_{-1g}	Push-down maneuver angle of attack	$[-20, 10]^\circ$	✓	✓
α_{cruise}	Cruise angle of attack	$[-10, 20]^\circ$		✓
θ	Twist	$[-15, 15]^\circ$		✓

(c) Initial Planform

The initial planform used to replicate this study is shown in the following figure 4

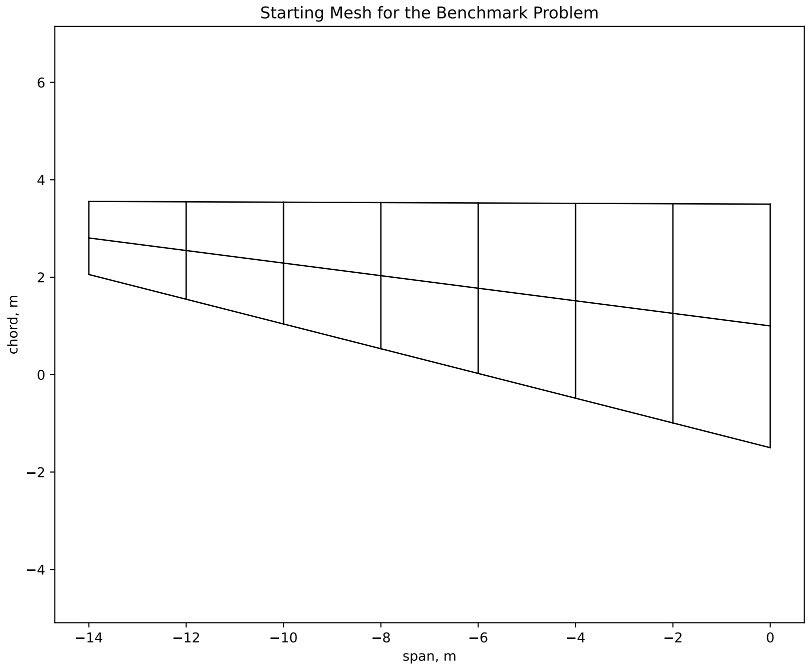


Figure 4: Starting Planform for the Benchmark Problems for this study

The specifications for the initial mesh are given as follows:

Specification	**Value**
Root Chord	5m
Tip Chord	1.5m
Span	28m
Twist	-
t/c	0.121
sweep	28.2 degrees

Table 5: Initial Planform Specifications

(d) Case (I) of the Benchmark Problem

The Case (I) is conducted to minimize the wingmass while only changing the structural panel thicknesses of the spars and skins. The optimal values of interest are given in the following table:

Optimal Parameter	**Value**
Fuelburn	8918.620 kg
Wingmass	1772.487 kg
Wingbox mass	558.697 kg

Table 6: Quantities of importance for Case 1

The wingbox mass is underpredicted by this model as compared to the 706kg obtained by Christison Gray and Martins (2024). The optimal skin and spar thickness values are shown below:

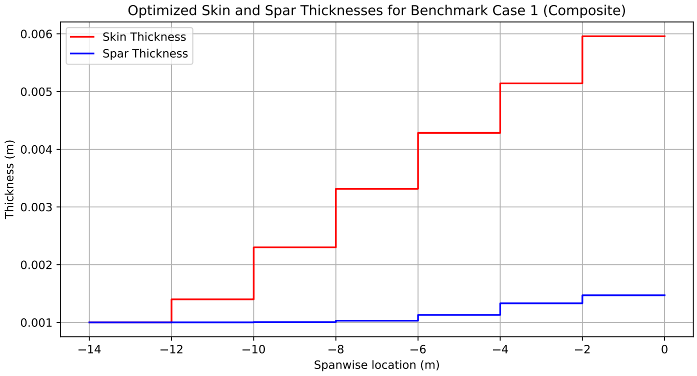
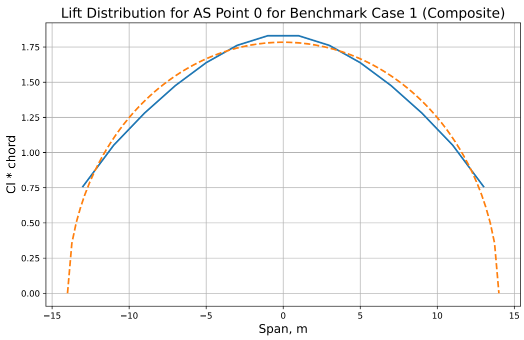
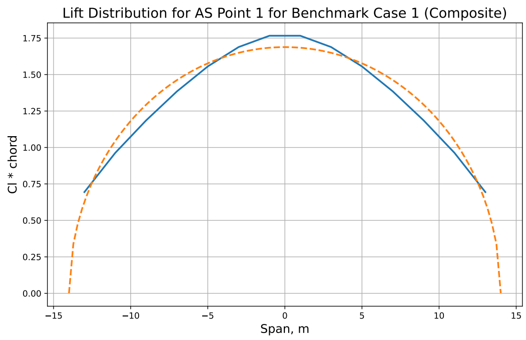


Figure 5: Optimized Skin and Spar panels' thickness distribution for Case 1

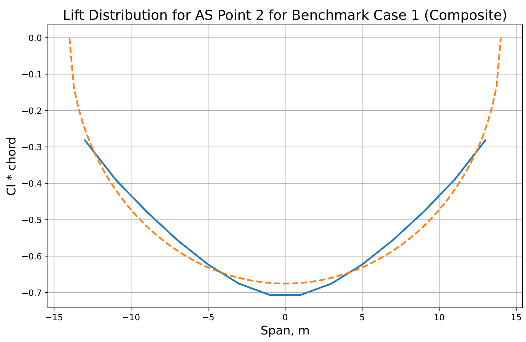
The lift distribution at each flight condition is shown in the following figure:



(a) Optimal Lift Distribution for Cruise Flight Condition



(b) Optimal Lift Distribution for Pull-up Maneuver Flight Condition



(c) Optimal Lift Distribution for Pull-down Maneuver Flight Condition

Figure 6: Optimal lift distribution at different flight points for Case 1

(e) Case (II) of the Benchmark Problem

The Case (II) is conducted to minimize the fuelburn while keeping the planform same, and only changing the structural panel thicknesses, along with the twist distribution and thickness to chord ratio distribution along the span of the wing. The values of interest at the optimal design are given in the following table:

Optimal Parameter	Value
Fuelburn	8346.911 kg
Wingmass	3718.787 kg
Wingbox mass	1385.048 kg

Table 7: Quantities of importance for Case 2

The fuel burn is right on point with the value obtained by Christison Gray and Martins (2024) at 8311 kgs. The optimized distribution of skin and spar panel thicknesses is given below:

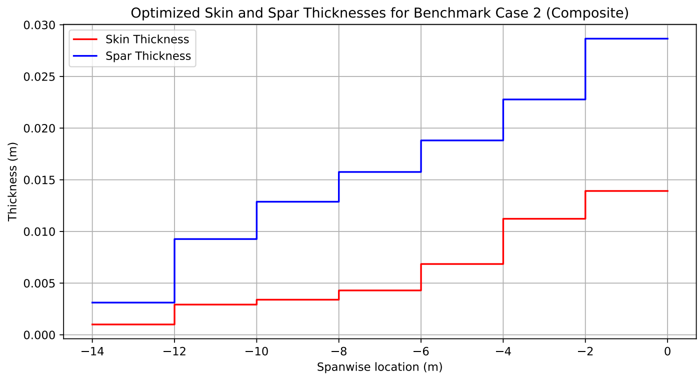


Figure 7: Optimized Skin and Spar panels' thickness distribution for Case 2

The Optimal Twist distribution is shown in the following figure:

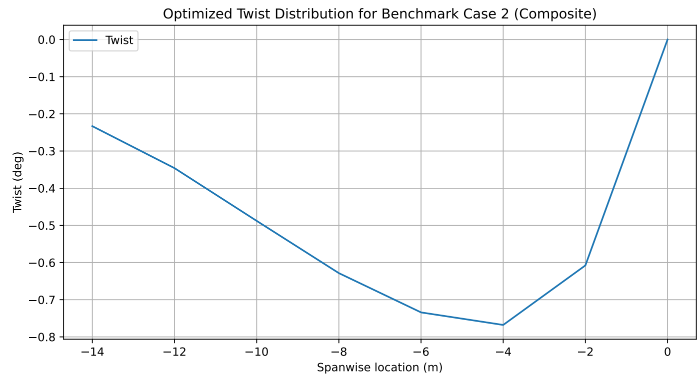


Figure 8: Optimized twist distribution for Case 2

The optimal thickness to chord ratio along the span of the optimal design is shown below:

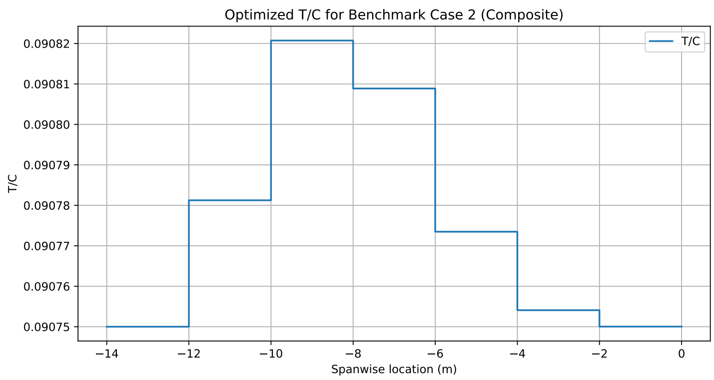
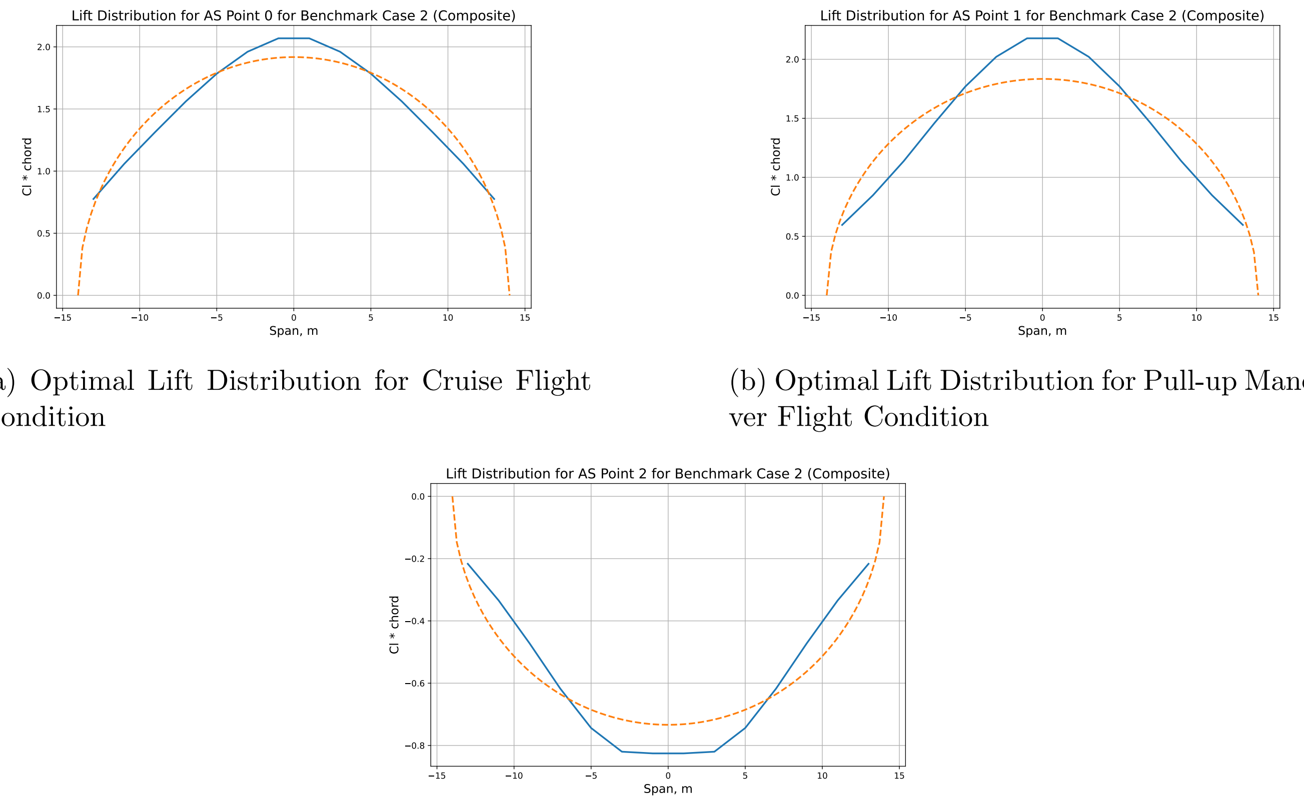


Figure 9: Optimized thickness to chord ratio distribution for Case 2

The spanwise lift distribution for the optimal design at each flight condition is shown in the following figure:



(a) Optimal Lift Distribution for Cruise Flight Condition

(b) Optimal Lift Distribution for Pull-up Maneuver Flight Condition

(c) Optimal Lift Distribution for Pull-down Maneuver Flight Condition

Figure 10: Optimal lift distribution at different flight points for Case 2

Conclusion

A Tsai-Wu based failure model for composite structures is developed for the OpenAeroStruct framework. The model is then utilized to solve the benchmark model developed by Christison Gray and Martins (2024). The current low fidelity model underpredicts the wingmass value obtained in Case (I). However, the model is right on point with the value of optimal fuelburn for the Case (II). The composite wingbox model shows promising results with the optimal thickness distributions of skin and spar panels, but varies slightly than what was reported by Christison Gray and Martins (2024).

References

Chauhan, S. and Martins, J. (2018). Low-fidelity aerostructural optimization of aircraft wings with a simplified wingbox model using openaerostruct. pages 418–431.

Christison Gray, A. and Martins, J. (2024). A proposed benchmark model for practical aeroelastic optimization of aircraft wings.